"اسئلة سنوات"

* find Domain, $f(x) = 3 + \sqrt{x-2}$.

sol:

$D_f = [2, \infty)$ $\quad x - 2 \geq 0$

$x \geq 2$

* $\lim_{x \to \infty} \frac{2x^3 - 4x + 1}{5x + 3x^3}$.

sol:-

$\lim \frac{2x^3}{3x^3} \Rightarrow \frac{2}{3}$ ✳

* The slope of the tangent line of the graph, $y = \cot\sqrt{x}$; $x = \frac{\pi^2}{4}$

sol:

$m = y' = \frac{-1}{2\sqrt{x}} \cdot \csc^2\sqrt{x}$

$= -\frac{1}{2 \cdot \frac{\pi}{2}} \cdot \csc^2\frac{\pi}{2} \Rightarrow -\frac{1}{\pi} \cdot 1 \Rightarrow -\frac{1}{\pi}$ ✳

* $f(x) = \tan^4 3x$, find $f'(x)$.

sol:

$f'(x) = 4\tan^3 3x \cdot 3 \cdot \sec^2 3x$

$f'(x) = 12 \tan^3 3x \cdot \sec^2 3x$

* If $x^2 - y^2 = 3x^2y^2$, Then dy/dx

sol:

$2x - 2yy' = 6xy^2 + 6x^2yy'$

$y'(-2y - 6x^2y) = 6xy^2 - 2x$

$y' = \frac{6xy^2 - 2x}{-2y - 6x^2y}$

* If $3x + \sin y = 4$ Then $\frac{d^2y}{dx^2}$

sol:

$3 + y'\cos y = 0$

$y'\cos y = -3 \rightarrow y' = \frac{-3}{\cos y}$

$y'' = \frac{-3 \cdot y' \cdot \sin y}{(\cos y)^2}$

$y'' = \frac{+3\sin y \cdot \frac{+3}{\cos y}}{(\cos y)^2}$ ✳

الممسوحة ضوئيا بـ CamScanner

* find the equation of the tangent line of the graph

$y = x \cos 3x$, at $x = \pi$

sol:

$m = y' = \cos 3x + (x)(3 - \sin 3x)$

$y' = \cos 3x - 3x \sin 3x$

$y' = \cos 3\pi - 3\pi \sin 3\pi \Rightarrow$ ($\sin 3\pi$ → zero)

$\boxed{m = y' = -1}$

$\boxed{m = -1}$ $\quad (\pi, -\pi)$

$y - y_0 = m(x - x_0)$

$y + \pi = -1(x - \pi)$

$y + \pi = -x + \pi$

$\boxed{y = -x}$ *

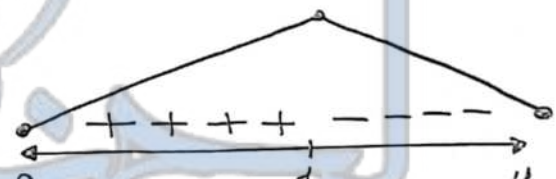
* find the absolute Min for, $f(x) = 2x - x^2$, on $[0, 4]$

sol:

$f'(x) = 2 - 2x = 0$

$2 = 2x$

$\boxed{x = 1}$



$f(0) = 0$ $\quad f(4) = -8$

absolute Min : $x = 4$

$f(4) = -8$

$(4, -8)$

* find Range , $f(x) = \sin x$

$R = [-1, 1]$

الممسوحة ضوئيا بـ CamScanner

* $f(x) = 3x^{\frac{2}{3}} - 2x$, find : a) increasing and decreasing
b) concavity
a) critical point and the esup
?

sol :-

$f'(x) = 2x^{-\frac{1}{3}} - 2$

$f'(x) = \frac{2}{\sqrt[3]{x}} - 2$

$f'(x) = \frac{2 - 2\sqrt[3]{x}}{\sqrt[3]{x}} = 0$

$2 = 2\sqrt[3]{x}$

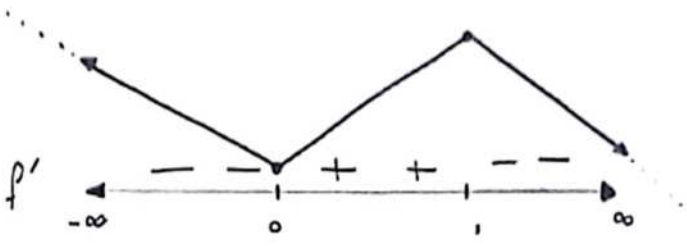
$(x^{\frac{1}{3}})^3 = (1)^3 \rightarrow \boxed{x = 1}$

$\boxed{x = 0}$

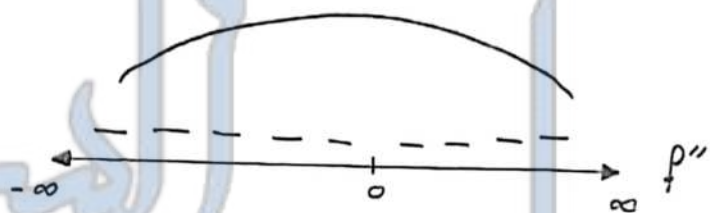
(1 , 1) (0 , 0)

$f''(x) = \frac{-2}{\sqrt[3]{(x)^2}} = 0$

$\boxed{x = 0}$



* f inc : $[0, 1]$
* f dec : $(-\infty, 0] \cup [1, \infty)$



f con up : None
f con down : $(-\infty, \infty)$
inflection points : None

* Find the Range of the function . $f(x) = \frac{1}{\sqrt{4 - x^2}}$

$y = \frac{1}{\sqrt{4 - x^2}}$

$x = \frac{1}{\sqrt{4 - y^2}}$

$\left(\sqrt{4 - y^2}\right)^2 = \left(\frac{1}{x}\right)^2$

$4 - y^2 = \frac{1}{x^2}$

$\sqrt{y^2} = \sqrt{4 - \frac{1}{x^2}}$

$y = \sqrt{4 - \frac{1}{x^2}}$

$\rightarrow f^{-1}(x) = \sqrt{4 - \frac{1}{x^2}}$

$\rightarrow f^{-1}(x) = \sqrt{\frac{4x^2 - 1}{x^2}}$

$\rightarrow f^{-1}(x) = \frac{\sqrt{4x^2 - 1}}{x}$

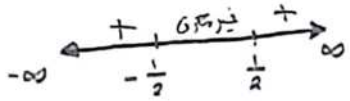
* $R = (-\infty, -\frac{1}{2}] \cup [\frac{1}{2}, \infty) - \{0\}$

$4x^2 - 1 \geq 0$

$4x^2 \geq 1$

$\sqrt{x^2} \geq \sqrt{\frac{1}{4}}$

$\boxed{x \geq \pm\frac{1}{2}}$



الممسوحة ضوئيا بـ CamScanner

* $f(x) = \frac{1}{x-3}$, $g(x) = \frac{1}{x}$, Then the domain $(f \circ g)(x)$

sol :-

$D_{(f \circ g)} = D_g \cap D_{f \circ g}$

$\mathbb{R} - \{0\} \cap \mathbb{R} - \{\frac{1}{3}\}$

$D_{(f \circ g)(x)} = \mathbb{R} - \{0, \frac{1}{3}\}$

$f(g(x)) = f(\frac{1}{x})$

$\Rightarrow f(\frac{1}{x}) = \frac{1}{\frac{1}{x} - \frac{3}{1}}$

$f(\frac{1}{x}) = \frac{1}{\frac{1-3x}{x}}$

$f(\frac{1}{x}) = \frac{x}{1-3x}$

---

* $y = f(x)$ domain $[2,3)$, find domain $g(x) = f(3-x)$

sol :

$2 \geq 3 - x > 3$

$-1 \geq -x > 0$

$1 \leq x < 0$

$D = [1, 0)$

---

* $f(x) = x^2$ and $g(x) = \begin{cases} 2x & , x < 4 \\ x+1 & , x \geq 4 \end{cases}$ Then $(f \circ g)(x)$

sol :

$(f \circ g)(x) = f(g(x))$

$= \begin{cases} 4x^2 & , x < 4 \\ (x+1)^2 & , x \geq 4 \end{cases}$

---

* $f(x) = \ln(5 - |7x+3|)$ is find Domain ?

sol :

$5 - |7x+3| > 0$

$5 > |7x+3|$

$-5 < 7x + 3 < 5$

$-3 \quad\quad -3$

$-8 < 7x < 2$

$-\frac{8}{7} < x < \frac{2}{7}$

Domain : $(-\frac{8}{7}, \frac{2}{7})$ ✳

الممسوحة ضوئيا بـ CamScanner

* The x-intercept of the function $f(x) = -2 + \ln(3x+1)$
  → $y = 0$

Sol :-

$$y = -2 + \ln(3x+1) \rightarrow \text{zero} = -2 + \ln(3x+1)$$

$$\rightarrow e^2 = e^{\ln(3x+1)} \rightarrow e^2 = 3x+1$$

$$\rightarrow 3x = e^2 - 1 \rightarrow x = \frac{e^2-1}{3}$$

* $\lim_{x \to -\infty} \left(\frac{\sqrt{49x^2+x+1} + 2x + 9}{6x+4} \right)$

Sol :

$$\lim \frac{\sqrt{49x^2} + 2x}{6x} \rightarrow \frac{-7x + 2x}{6x} \quad -\frac{5}{6} \quad *$$

* $\lim_{x \to 2^-} \left(\frac{1}{x-2} - \frac{12}{x^3-8} \right)$

$x^3 - 8 = (x-2)(x^2+2x+4)$

Sol :

$$= \frac{0}{0}$$

$$\lim_{x \to 2^-} \left(\frac{x^2+2x+4}{x^3-8} - \frac{12}{x^3-8} \right)$$

$$\lim_{x \to 2^-} \frac{x^2+2x-8}{(x^3-8)} \Rightarrow \frac{\cancel{(x-2)}(x+4)}{\cancel{(x-2)}(x^2+2x+4)}$$

$$\lim_{x \to 2^-} \frac{x+4}{x^2+2x+4} \Rightarrow \frac{1}{2} \quad *$$

الممسوحة ضوئيا بـ CamScanner

* The x-intercept of the function $f(x) = -2 + \ln(3x+1)$
  → y = 0

sol :-

$$y = -2 + \ln(3x+1) \rightarrow \text{zero} = -2 + \ln(3x+1)$$

$$\rightarrow e^2 = e^{\ln(3x+1)} \rightarrow e^2 = 3x+1$$

$$\rightarrow 3x = e^2 - 1 \rightarrow x = \frac{e^2-1}{3}$$

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sol :

$$\lim \frac{\sqrt{49x^2}+2x}{6x} \rightarrow \frac{-7x+2x}{6x} \quad \frac{-5}{6} \ *$$

* $\lim_{x \to 2^-} \left(\frac{1}{x-2} - \frac{12}{x^3-8} \right)$

$x^3 - 8 = (x-2)(x^2+2x+4)$

sol :

$$= \frac{0}{0}$$

$$\lim_{x \to 2^-} \left(\frac{x^2+2x+4}{x^3-8} - \frac{12}{x^3-8} \right)$$

$$\lim_{x \to 2^-} \frac{x^2+2x-8}{(x^3-8)} \Rightarrow \frac{(x-2)(x+4)}{(x-2)(x^2+2x+4)}$$

$$\lim_{x \to 2^-} \frac{x+4}{x^2+2x+4} \Rightarrow \frac{1}{2} \ *$$

الممسوحة ضوئيا بـ CamScanner

* $f(x) = \begin{cases} 2 - x & , \ x \le 0 \\ 1 - e^{-x} & , \ x \ge 0 \end{cases}$ Then $f^{-1}(x) = ?$

Sol :-

$$f^{-1}(x) = \begin{cases} 2 - x & , \ x > 2 \\ -\ln(1-x) & , \ 1 < x \le 2 \end{cases}$$

"اللهم إني استودعك ما حفظت
وما قرأت وما درست فردّه إليّ عند حاجتي
إليه يارب العالمين"

"أمي تعلم أمي تعلم أمي ... تعلم أبي
تعلم لا أحد"

الممسوحة ضوئيا بـ CamScanner

* If $f(3x) = 9x + 5$, Then $f(x)$.

Sol:

$3x = y$

$\boxed{x = \frac{y}{3}}$

$\rightarrow f(y) = 9 \cdot (\frac{y}{3}) + 5$

$\rightarrow f(y) = 3y + 5$

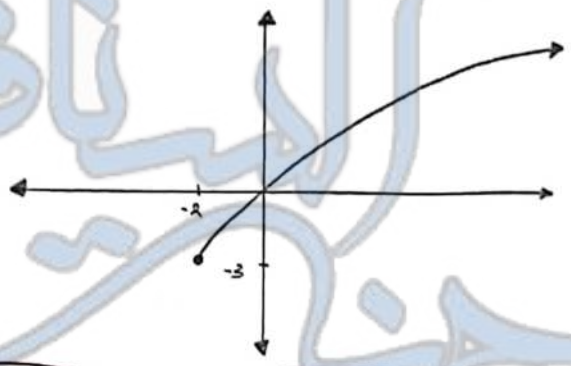
$\rightarrow f(x) = 3x + 5$ *

* The function $f(x) = \frac{x^5 |x|}{x^4 - 3}$, even or odd

Sol:

$$f(-x) = \frac{(-x)^5 |-x|}{(-x)^4 - 3} \Rightarrow \frac{-x^5 |x|}{x^4 - 3}$$ odd

* The graph of $f(x) = \sqrt{x + 2} - 3$



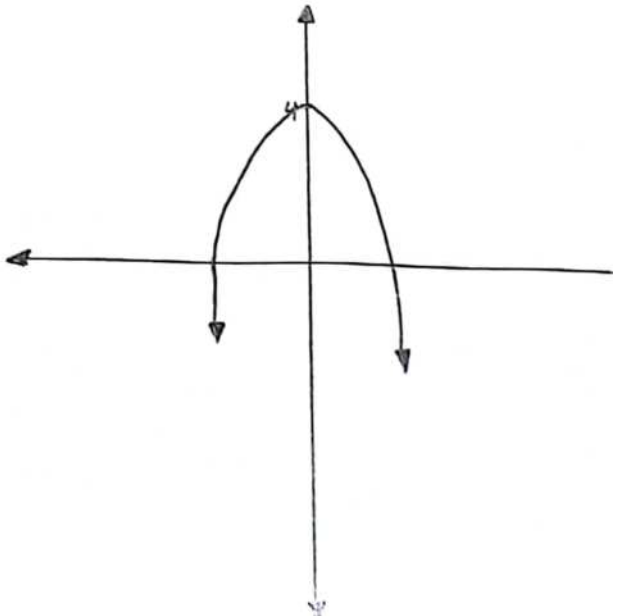
* The graph of $f(x) = -x^2 + 8x - 16$

Sol:

$\frac{-b}{2a} \rightarrow \frac{-8}{-2} \rightarrow 4$

$f(\frac{-b}{2a}) \rightarrow f(4) = 0$

R : $(-\infty, 0)$



الممسوحة ضوئيا بـ CamScanner

* $f(x) = \sqrt{x-1}$, $g(x) = \frac{1}{x^2-5}$ Then $D_{g\circ f}$

Sol:

$D_{g\circ f} = D_f \cap D_{g\circ f}$

$= [1,\infty) \cap \mathbb{R}-\{6\}$

$D_{g\circ f} = [1,\infty)-\{6\}$

$(g\circ f)(x) \rightarrow g(f(x))$

$\rightarrow g(\sqrt{x-1}) \rightarrow \frac{1}{(\sqrt{x-1})^2-5}$

$\rightarrow \frac{1}{x-1-5} \rightarrow \frac{1}{x-6}$

---

* $f(x) = 2x^3 - 3x^2 + 1$, $f^{-1}(x) = 1$ Then x =

Sol:

$y = 2x^3 - 3x^2 + 1$

$x = 2y^3 - 3y^2 + 1$

$x = 2 - 3 + 1 \Rightarrow$ Zero

---

* $\log_x (2x+3) = 2$, Then x

Sol :-

$= x^2 = 2x + 3 \rightarrow x^2 - 2x - 3 =$ Zero

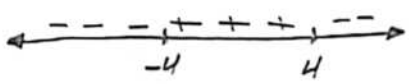
$x = 3$, $x = -1$ ✳

---

* find Domain of $f(x) = 1 + \sqrt{16 - x^2}$

$16 - x^2 \geq 0$

$16 \geq x^2 \rightarrow x \leq \pm 4$



$D_f = [-4, 4]$

---

* The Domain and Rang $f(x)$ are $[-1,3)$ and $(0,4)$ respectively, Then the domain of $g(x) = 1 - f(3-2x)$

Sol:

$-1 \geq 3 - 2x > 3$

$-4 \geq -2x > 0$ $\quad D_g = [2, 0)$

$2 \leq x < 0$

الممسوحة ضوئيا بـ CamScanner

* The Rang of $f(x) = \sin^{-1}(3x+1) + \frac{\pi}{2}$

sol:

Rang $\sin^{-1}(x) = [-\frac{\pi}{2}, \frac{\pi}{2}]$

$$-\frac{\pi}{2} \leq \sin^{-1}(3x+1) + \frac{\pi}{2} \leq \frac{\pi}{2}$$

$$-\frac{\pi}{2} \leq \sin^{-1}(3x+1) \leq 0$$

$$R\left(\sin^{-1}(3x+1) + \frac{\pi}{2}\right) = [0, \pi]$$

---

* $\sin^{-1}\left(\sin \frac{6\pi}{7}\right) = \frac{6\pi}{7}$

---

* $f(x) = \frac{e^x}{1-5e^x}$. find .

① D_f , ② $f^{-1}(x)$ ③ Rang f

sol:

① $D_f = R - \{\ln \frac{1}{5}\}$

② $y = \frac{e^x}{1-5e^x} \rightarrow x = \frac{e^y}{1-5e^y} \rightarrow x - 5xe^y = e^y$

$$x = e^y(1+5x) \rightarrow \ln e^y = \ln \frac{x}{1+5x}$$

$$y = \ln\left(\frac{x}{1+5x}\right) \rightarrow f^{-1}(x) = \ln\left(\frac{x}{1+5x}\right)$$

③ Rang f = Domain f^{-1}

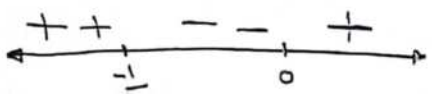
$D_{f^{-1}} = (0, \infty), (-\infty, -\frac{1}{5})$

$\frac{x}{1+5x} > 0$

$x > 0 \rightarrow (0, \infty)$

$1+5x > 0$

$5x > -1 \rightarrow [-\frac{1}{5}, \infty)$



الممسوحة ضوئيا بـ CamScanner

* evaluate The limit :

$$\lim_{x \to 2} \frac{1}{4x^2 - 16}$$

sol :-

$$\lim_{x \to 2^+} \frac{1}{4x^2 - 16} \quad , \quad \lim_{x \to 2^-} \frac{1}{4x^2 - 16}$$

$$= \frac{1}{0} \to \infty \quad , \quad = \frac{1}{0} \to -\infty$$

d.n . e

* find The value of "x" , $\sin(3x) = 0$ و $x \in [0, \frac{\pi}{3}]$

sol :

$$3x = \sin^{-1}(0)$$

$$3x = 0, \pi, 2\pi$$

$$x = 0, \frac{\pi}{3}, \frac{2\pi}{3}$$

تحل
لانها خارج المجال

$x = 0$ $\quad$ $x = \frac{\pi}{3}$

* find The value of "x" , $\ln(x-1) = 2$

sol :

$$e^{\ln(x-1)} = e^0 \quad \to \quad x - 1 = 1$$

$$x = 2$$

* evaluate The limit ; $\lim_{x \to 0} \frac{\sin 3x^6}{\tan^2(x^3)}$

sol :

$$\lim_{x \to 0} \frac{\sin 3x^6}{\tan^2(x^3)} \to \frac{3x^6}{x^6} \to 3$$

الممسوحة ضوئيا بـ CamScanner

* evaluate the limit ; $\lim_{x \to 3} \frac{\sqrt{x+1} - 2}{x - 3}$

soll:

$$\lim_{x \to 3} \frac{\sqrt{x+1} - 2}{x - 3} \to \frac{0}{0}$$

$$\xrightarrow{L.H} \lim_{x \to 3} \frac{\frac{1}{2\sqrt{x+1}}}{1} \to \frac{1}{4} \ **$$

---

* study the continuity of the function , $f(x) = \ln(x^2 + e^x)$.

soll:

$$\lim_{x \to \infty} \ln(x^2 + e^x) \quad , \quad \lim_{x \to -\infty} \ln(x^2 + e^x)$$

$$\ln(\infty) \quad , \quad \ln(\infty)$$

$$\infty \quad , \quad \infty$$

continuty every where

---

* study the continuity of the function at $x = 0$.

$$f(x) = \begin{cases} e^{2x} - 1 & , \ x > 0 \\ 0 & , \ x \leq 0 \end{cases}$$

$$\lim_{x \to 0^+} e^{2x} - 1 \quad \stackrel{?}{=} \quad \lim_{x \to 0^-} 0$$

$$0 \quad = \quad 0$$

The function is continuity at $\boxed{x = 0}$

الممسوحة ضوئيا بـ CamScanner

* Find Domain of $f(x) = \frac{\cos x}{|x| - 4}$.

sol:

$D_f = \mathbb{R} - \{\pm 4\}$

$|x| - 4 = 0$

$|x| = 4$

$x = \pm 4$

---

* Find Domain of $f(x) = \ln(1 + \ln x)$.

sol:

$x > 0$, $(0, \infty)$

$1 + \ln x > 0$

$\ln x > -1$

$e^{\ln x} > e^{-1}$

$\boxed{x > \frac{1}{e}}$

$D_f = (0, \infty)$

---

* $f(x) = \ln(1 + \ln x)$ Then $f^{-1}(\ln 2)$

sol:

$y = \ln(1 + \ln x) \rightarrow x = \ln(1 + \ln y)$

$\rightarrow e^x = 1 + \ln y \rightarrow e^{\ln y} = e^{(e^x - 1)}$

$\rightarrow y = e^{(e^x - 1)} \rightarrow f^{-1}(x) = e^{(e^x - 1)}$

$\rightarrow f^{-1}(\ln 2) = e^{(e^{\ln 2} - 1)} \rightarrow f^{-1}(\ln 2) = e^{2-1}$

$f^{-1}(\ln 2) = e$

الألم يخير الناس

فيجعلهم يتفون أقل

ويفكرون أكثر و ينحزلون أطول"

الممسوحة ضوئيا بـ CamScanner

* Find Range $f(x) = \frac{4}{4 + 2\sin^3 x}$

sol:-

$\rightarrow (-1 \geq \sin x \geq 1)^3$

$\rightarrow \quad -1 \geq \sin^3 x \geq 1 \longrightarrow \quad -2 \geq 2\sin^3 x \geq 2$

$\rightarrow \quad 2 \geq 4 + 2\sin^3 x \geq 6$

$\frac{1}{2} \geq \frac{1}{4 + 2\sin^3 x} \geq \frac{1}{6}$

$\rightarrow \quad 2 \geq \frac{4}{4 + 2\sin^3 x} \geq \frac{2}{3}$

$\therefore R_f = [\frac{2}{3}, 2]$

* $\cot^2(\tan^{-1}(x))$

sol:

$\frac{1}{(\tan(\tan^{-1}(x)))^2} \rightarrow \frac{1}{x^2}$ *

* $f(x) = 2x + 1$ and $f(g(x)) = x^3 + x^2 + x + 1$

then $g(-1) = ?$

$(f \circ g)(x) = f(g(x))$

$x^3 + x^2 + x + 1 = 2g(x) + 1$

$-1 + 1 - 1 + 1$

$0 = 2g(-1) + 1$

$-1 = 2g(-1) \longrightarrow g(-1) = -\frac{1}{2}$

الممسوحة ضوئيا بـ CamScanner

* Find Domain of $f(x)$

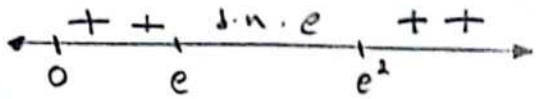
$$f(x) = \sqrt{(\ln x)^2 - 3\ln x + 2}$$

sol:

$$(\ln x)^2 - 3\ln x + 2 \geq 0$$

$\ln x = 2$ $\quad$ $\ln x = 1$

$\boxed{x = e^2}$ $\quad$ $\boxed{x = e}$



$$D_f = (0, e] \cup [e^2, \infty)$$

* Find Domain of $f(x) = \ln(x+1) + \sqrt{4 - x^2}$

sol:

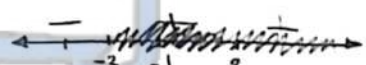
$$f(x) = \ln(x+1) + \sqrt{4 - x^2}$$

$D = (-1, \infty) \cap [-2, 2]$

$4 \geq x^2$

$\boxed{x \leq \pm 2}$

$$D_f = (-1, 2]$$



* $\lim_{x \to 0} \frac{\sin^2 3x}{3x^2}$

sol:

$$\lim_{x \to 0} \frac{\sin^2 3x}{3x^2} \Rightarrow \frac{9x^2}{3x^2} \to 3$$

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x \to \infty} \sqrt{4x^2 + x} - 2x$

sol:

$\sqrt{4(\infty) + \infty} - 2 \cdot \infty \rightarrow \infty - \infty$

$\Rightarrow$ Zero

* $\lim_{x \to 0^-} \frac{x^3 - 1}{x^4 - x^3} \rightarrow \frac{-1}{0} \rightarrow -\infty$

* The number of discontinuities of the function.

$$f(x) = \frac{13 - x}{5x^2 + 3\sin^6 x}$$

??

* $f(x) = \frac{x+1}{x-1}$, $x > 1$ Then $f^{-1}(x)$

sol:

$$y = \frac{x+1}{x-1} \rightarrow x = \frac{y+1}{y-1} \rightarrow xy - x = y + 1$$

$$\rightarrow y(x-1) = 1 + x$$

$$y = \frac{1+x}{x-1} \Rightarrow \therefore f^{-1}(x) = \frac{1+x}{x-1}$$ ##

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x \to 2} \frac{\sqrt{x^2+12} - 4}{x - 2}$

sol: $\lim_{x \to 2} \frac{\sqrt{x^2+12} - 4}{x - 2} = \frac{0}{0}$

$\lim_{x \to 2} \frac{\frac{2x}{2\sqrt{x^2+12}}}{1} \Rightarrow \frac{2}{4} \to \frac{1}{2}$ *

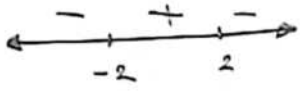
* $\lim_{x \to -1} \frac{\tan(x-1)}{x^2 - 1}$

sol: $\lim_{x \to -1} \frac{\tan(x-1)}{x^2 - 1} \Rightarrow \frac{0}{0}$

$\lim_{x \to -1} \frac{\sec^2(x-1)}{2x} \Rightarrow \frac{1}{-2}$ *

* Find Domain function, $f(x) = \cos\left(\frac{1}{\sqrt{4-x^2}}\right)$ is.

sol: $4 - x^2 \geq 0 \to 4 \geq x^2 \to \boxed{x = \pm 2}$



$D = (-2, 2)$??

* Find Range function, $f(x) = 2 - \sqrt{x-2}$

$R = (-\infty, 2]$

"النجاح سلم لا تستطيع تسلقه، ويداك في جيبك"

الممسوحة ضوئيا بـ CamScanner

* If $\ln(\ln(x-1)) = \ln 2 - \ln 3$ Find "x"

sol:

$\rightarrow \ln(\ln(x-1)) = \ln(\frac{2}{3})$

$\rightarrow e^{\ln x-1} = e^{\frac{2}{3}} \rightarrow x-1 = e^{\frac{2}{3}}$

$\rightarrow x = \sqrt[3]{e^2} + 1$ ※

* $\lim_{x\to 0} \frac{\sin^9(4x)}{x^8}$

sol:

$\lim_{x\to 0} \frac{\sin^9(4x)}{x^8} = \frac{0}{0} \rightarrow \frac{(4x)^9}{x^8} \rightarrow \frac{262144\,x^9}{x^8}$.

$\Rightarrow 262144\,x \rightarrow Zero$

* $\lim_{x\to\infty} [\sqrt{x^2+(a+2)x} - \sqrt{x^2+2x}] = \frac{a^9}{2}$

sol:

$\lim_{x\to\infty} [\sqrt{x^2} - \sqrt{x^2}] = \frac{a^9}{2}$

$0 = \frac{a^9}{2}$

$a^9 = 0 \rightarrow$ عيادة $\boxed{0 = \ln(1)}$

$e^{a^9} = e^{\ln 1}$

$\boxed{a = 1}$

الممسوحة ضوئيا بـ CamScanner

* The H.A For , $f(x) = e^{2-x^2}$.

$$\lim_{x\to\infty} e^{2-x^2} \to e^{-\infty} \to \text{zero}$$

$$\lim_{x\to-\infty} e^{2-x^2} \to e^{\infty} \to \infty$$

$\Big\}$ $y = 0$

* $$\lim_{x\to\infty} \frac{2x+1}{\sqrt{3x^2+5}-7}$$

sol:

$$\lim_{x\to\infty} \frac{2x}{\sqrt{3}x} \to \frac{2}{\sqrt{3}} \#$$

* $$\lim_{x\to 0} \frac{\sin(2x)}{x+\tan(3x)}$$

sol:

$$\lim_{x\to 0} \frac{\sin 2x}{x+\tan 3x} = \frac{0}{0}$$

L.H $\Rightarrow$ $$\lim_{x\to 0} \frac{2\cos 2x}{1+3\sec 3x} \to \frac{2(1)}{1+3(1)^2} \to \frac{1}{2} \#$$

* $$\lim_{x\to-\infty} \frac{3x-5x^3+1}{x-x^5+1}$$

sol:

$$\lim_{x\to-\infty} \frac{+5x^3}{+x^5} \to \frac{5}{\infty} \to \text{zero} \#$$

قال الله تعالى :

"قل يا عبادي الذين أسرفوا على أنفسهم لا تقنطوا من رحمة الله إن الله يغفر الذنوب جميعا انه هو الغفور الرحيم"

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x \to 0} x^2 \left[\sin^2\left(\cos \frac{5}{x}\right) + \frac{2}{\sin^2(2x)} \right]$

sol:-

$$\lim_{x \to 0} x^2 \cdot \sin^2\left(\cos \frac{5}{x}\right) + \lim_{x \to 0} \frac{2x^2}{\sin^2(2x)}$$

$$0 \leq \sin^2\left(\cos \frac{5}{x}\right) \leq 1$$

$$0 \leq x^2 \cdot \sin^2\left(\cos \frac{5}{x}\right) < x^2$$

$0 \to$ zero, $x^2 \to$ zero, $\sin^2\left(\cos \frac{5}{x}\right) \to$ zero

$$\Rightarrow \frac{2x^2}{4x^2} \to \frac{1}{2}$$

$$= 0 + \frac{1}{2} = 1/2 \quad \#$$

* find V.A of the function $f(x) = \frac{2}{x} + \frac{1}{x^2}$?

sol:

$$f(x) = \frac{2x+1}{x^2}$$

$x = 0$ V.A

* $f(x) = \sqrt{x^2 - x} - x^{\mp}$ find number of H.A ?

sol:

$$\lim_{x \to \infty} \sqrt{x^2 - x} - x^{\mp} = \sqrt{x^2} - x^{\mp} \to \infty - \infty = \mp$$

$$\lim_{x \to -\infty} \sqrt{x^2 - x} - x^{\mp} = \sqrt{x^2} - x^{\mp} \to -\infty + \infty = \mp$$

H.A = $\mp$

اصنع حياتك ولا تستنسخ حياة الآخرين

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x \to 1} \frac{e^{2\ln x} - 9}{x^3 - 3x^2 + x - 3}$

sol:

$$\lim_{x \to 1} \frac{e^{2\ln x^2} - 9}{x^3 - 3x^2 + x - 3} \Rightarrow -\frac{8}{-4} \Rightarrow 2$$

* $\lim_{x \to 2^-} \left(\frac{x+2}{2-x} \right)$

sol:

$$\lim_{x \to 2^-} \frac{x+2}{2-x} \to \frac{4}{0} \to \infty$$

* $\lim_{x \to \infty} \sqrt{x^2 - 5} - x$

sol

$$\lim_{x \to \infty} \sqrt{\infty - 5} - \infty = \infty - \infty \Rightarrow \text{Zero}$$

* $\lim_{x \to \infty} \sqrt{x^2 + 4x} - \sqrt{x^2 + x}$??

sol:

$$\lim_{x \to \infty} \sqrt{x^2} - \sqrt{x^2} \to \lim_{x \to \infty} x - x = 0$$

* $f(x) = \frac{x^2 - 1}{x^2 - 2x + 1}$, The H.A and V.A are?

sol:

V.A: $x^2 - 2x + 1 = 0$; H.A: $\lim_{x \to \pm\infty} \frac{x^2 - 1}{x^2 - 2x + 1} \Rightarrow \frac{(x-1)(x+1)}{(x-1)^2}$

$$\boxed{x = 1}$$

$$\lim_{x \to \infty} \frac{x-1}{x+1} \to \frac{\infty}{\infty} \to 1$$

$$\lim_{x \to -\infty} \frac{x-1}{x+1} \to \frac{\infty}{\infty} \to 1$$

$$\boxed{y = 1}$$

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x\to\infty} \sqrt{x+2\sqrt{x}} - \sqrt{x}$

sol:

$$\lim_{x\to\infty} \sqrt{x+3\sqrt{x}} - \sqrt{x} \quad \to \quad \infty - \infty$$

$$\lim_{x\to\infty} \sqrt{x+3\sqrt{x}} - \sqrt{x} \cdot \frac{\sqrt{x+3\sqrt{x}} + \sqrt{x}}{\sqrt{x+3\sqrt{x}} + \sqrt{x}}$$

$$\lim_{x\to\infty} \frac{x+3\sqrt{x} - x}{\sqrt{x+3\sqrt{x}} + \sqrt{x}} \to \frac{3\sqrt{x}}{\sqrt{x}\left(\sqrt{1+\frac{3}{\sqrt{x}}} + 1\right)}$$

$$\lim \frac{3}{1+1} \to \frac{3}{2} \quad *$$

* $\lim \frac{\sqrt{9x+b} - 2}{x} = 4$ then the values of a and b are?

sol :-

$$\sqrt{9x+b} - 2 = a \quad , \quad \boxed{x = 0}$$

$$\sqrt{b} - 2 = 0$$

$$\sqrt{b} = 2 \to \boxed{b = 4}$$

??

* $\lim_{x\to\infty} \frac{3x^2 + \cos^2 x}{x^2 + 1}$

sol :→

$$\lim_{x\to\infty} \frac{\frac{3x^2}{x^2} + \frac{\cos^2 x}{x^2}}{\frac{x^2}{x^2} + \frac{1}{x^2}} \Rightarrow \frac{3+a}{1+a} = 3$$

الممسوحة ضوئيا بـ CamScanner

* The H.A of $f(x) = \sqrt{x^2+x+1} - \sqrt{x^2-4x}$

soll:-

$$\lim_{x \to \infty} \sqrt{x^2+x+1} - \sqrt{x^2-4x} \cdot \frac{\sqrt{x^2+x+1} + \sqrt{x^2-4x}}{\sqrt{x^2+x+1} + \sqrt{x^2-4x}}$$

$$\lim_{x \to \infty} \frac{x^2+x+1 - x^2+4x}{\sqrt{x^2+x+1} + \sqrt{x^2-4x}} \Rightarrow \frac{5x+1}{\sqrt{x^2}+\sqrt{x}}$$

$$\lim_{x \to \infty} \frac{5x+1}{x+x} \to \frac{5}{2}$$

$$\lim_{x \to -\infty} \frac{5x+1}{-x-x} \to -\frac{5}{2}$$

* $$\lim \frac{x + \sin 5x}{5x + \sin x}$$

soll:

$$\lim \frac{x+\square}{5x+\square} \Rightarrow \frac{1}{5}$$

* find the V.A of, $f(x) = \ln(x-2)$

soll:

$$x - 2 = \text{zero} \Rightarrow \boxed{x = 2}$$

استعن بالله ولا تعجز

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x \to \frac{3\pi}{2}^+} \frac{|\sin x|}{\sin x}$

sol:

$$\lim_{x \to \frac{3\pi}{2}^+} \frac{-\sin x}{\sin x} \to \lim \frac{-(-1)}{-1} \to -1$$

---

* $\lim_{x \to 2^+} (x-2) \cdot \sin\left(\frac{1}{x^2-4}\right)$

sol:

$\lim (0) \cdot (\text{d.n.e})$

$\lim (x-2) \cdot \left(\sin\left(\frac{1}{x^2-4}\right)\right)$

$= \text{Zero}$

$$(-1 \geq \sin\tfrac{1}{x^2-4} \geq 1)^{x-2}$$

$$-x+2 \geq (x-2)\sin\tfrac{1}{x^2-4} \geq x-2$$

$0 \qquad\qquad\qquad\qquad 0$

0

---

* find V.A of function $f(x) = \frac{2x^2+1}{(2x-1)^2-1}$

sol:

$$(2x-1)^2 - 1 = 0 \Rightarrow \sqrt{(2x-1)^2} = \sqrt{1}$$

$2x - 1 = -1$, $2x - 1 = 1$

$2x = 0$, $2x = 2$

$\boxed{x = 0}$, $\boxed{x = 1}$

---

* $\lim_{x \to 1} \frac{x^3 + 2x^2 - x + a}{x^2 - 1}$

sol:

$$x^3 + 2x^2 - x + a = 0$$

$$x + 2 - 1 + a = 0$$

$$2 + a = 0 \Rightarrow \boxed{a = -2}$$

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x \to 0} \frac{\sqrt{x+9} - 3}{x} = \frac{0}{0}$

sol:

$\lim_{x \to 0} \frac{\frac{1}{2\sqrt{x+9}}}{1} \Rightarrow \frac{1}{6}$

---

* $\lim_{x \to 0^-} 3x \cdot \sin\left(\frac{2}{x}\right)$

sol:

$-1 < \sin \frac{2}{x} < 1$

$-3x < 3x \cdot \sin\frac{2}{x} < 3x$

$\lim_{x \to 0^-} 3x \cdot \sin \frac{2}{x} = \text{zero}$

$\lim_{x \to 0^-} -3x = 0$, $\lim_{x \to 0^-} 3x = 0$, zero

---

* $\lim_{x \to 3^-} \frac{x^2 - 3x}{x^2 - 6x + 9}$

sol:

$\lim_{x \to 3^-} \frac{x^2 - 3x}{x^2 - 6x + 9} = \frac{0}{0} \Rightarrow \lim_{x \to 3^-} \frac{2x - 3}{2x - 6} = \frac{3}{0^-} = -\infty$

---

* $\lim_{x \to 3} \frac{\sin(x-3)}{x^2 - 2x - 3}$

sol:

$\lim_{x \to 3} \frac{\sin(x-3)}{x^2 - 2x - 3} = \frac{0}{0} \Rightarrow \frac{\cos(x-3)}{2x-2} = \frac{1}{4}$

$\lim_{x \to \infty} 3x \cdot \tan \frac{2}{x}$

sol: $\lim \frac{\tan \frac{2}{x}}{\frac{1}{3x}} \Rightarrow \frac{\frac{2}{x}}{\frac{1}{3x}} \to 6$

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x \to 0} \cos\left(\frac{\sin(x+\pi)}{2x}\right)$

sol:-

$\lim \cos\left(\frac{\sin(x+\pi)}{2x}\right) \rightarrow \lim \cos\left(\frac{0}{0}\right)$

$\lim \cos\left(\frac{\cos(x+\pi)}{2}\right) \rightarrow \left(\cos\left(\frac{1}{2}\right)\right)$ ✳

---

* $\lim_{x \to \infty} \left[\ln(3e^x - e^{-x}) - \ln(e^x - e^{-x})\right]$

sol:-

$\lim_{x \to \infty} \left[\ln\left(\frac{3e^x - e^{-x}}{e^x - e^{-x}}\right)\right] \rightarrow \ln\left(\frac{3e^x}{e^x}\right)$

$\rightarrow \ln(3)$

---

* The continuity points for, $f(x) = \frac{\ln(x-1)}{x^2 + x + 1}$

sol

$x - 1 > 0$

$x > 1 \quad D_f = (1, \infty)$

---

* $f(x) = \begin{cases} \frac{\tan |k| x}{x} & , x < 0 \\ 2x + 3|k|^2 & , x \geq 0 \end{cases}$

all values of $|k|$ That Let $\lim f(x)$ exists are

$\lim_{x \to 0} \frac{\tan |k| x}{x} = \lim_{x \to 0^+} 2x + 3|k|^2$

$\frac{|k|}{1} = 3|k|^2 \rightarrow \boxed{|k| = \frac{1}{3}}$

"من يخاف الموت يختصر الحياة"

الممسوحة ضوئيا بـ CamScanner

* $\lim_{x \to 0} \tan\left(\frac{1-\cos x}{x}\right)$

sol :-

$\lim_{x \to 0} \tan\left(\frac{0}{0}\right) \longrightarrow \lim \tan\left(\frac{\sin x}{\quad}\right)$

$\longrightarrow \lim \tan(0) \Rightarrow$ zero

* let $f(x) = \sqrt{|x| - 4}$ Then f continuous on ?

sol:

$|x| - 4 \geq 0$ or $x \geq 4$, $x \leq -4$

$|x| \geq 4$

$[4, \infty)$, $[-\infty, -4]$

* find values of "a" . $f(x) = \begin{cases} x + 4 & , x \leq a \\ x^2 & , x > a \end{cases}$

$\lim_{x \to a^-} f(x) = \lim_{a \to a^+} f(x)$

$a + 4 = a^2$

$a^2 - a - 4 = 0$ → $a = 2.56$, $a = -1.56$

* $f(x) = \frac{\csc x}{|x-1| - 2}$ is discontinuous at .

sol:

$|x - 1| = 2$

$x - 1 = 2$ $\quad x - 1 = -2$

$x = 3$ $\quad x = -1$

$\sin x = 0$

$x = 0$. $x = \pi$, x

الجنان
الممسوحة ضوئيا بـ CamScanner

* $f(x) = x^2 - 1$, $g(x) = 2x + 1$ Find $(f \circ g)(x)$

sol:

$(f \circ g)(x) = f(g(x)) \Rightarrow f(2x+1)$

$\Rightarrow (2x+1)^2 - 1$

---

* Find Domain of $f(x) = \sqrt{4 - x^2} + \frac{1}{x-2}$

sol:

$4 - x^2 \geq 0$

$4 \geq x^2 \rightarrow x \leq \pm 2$



$x - 2 = 0$

$x = 2$

$D = [-2, 2)$

---

* find $(f \circ g \circ h)(x)$: $f(x) = x + 1$, $g(x) = 2x$, $h(x) = x - 1$

sol:

$(f \circ g \circ h)(x) = f(g(h(x)))$

$= f(g(x-1)) \rightarrow f(2x-2)$

$\Rightarrow 2x - 2 + 1 \Rightarrow 2x - 1$

---

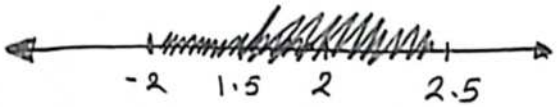
* $h(x) = \sqrt{4 - x^2} \cdot \sin^{-1}(2x - 4)$ Find Domain ??

$[-2, 2] \cap [1.5, 2.5]$

$-1 > 2x - 4 > 1$

$3 > 2x > 5$

$1.5 > x > 2.5$



$D_h = [1.5, 2]$

وانه لجهاد نصر أو استشهاد

وعبدة الناطق الرسمي باسم
كتائب عزدين القسام

الممسوحة ضوئيا بـ CamScanner

* If $f(x) = \frac{\cos^{-1}(1-x)}{2-x}$, find Domain is:

sol:

$$-1 > 1 - x > 1$$
$$-2 > -x > 0$$
$$2 < x < 0$$

$D_f = [0, 2)$

* let "x" be a number in the interval $[-1, 1]$ Then $\csc(\sin^{-1}(x))$ is:

sol:

$$\frac{1}{\sin(\sin^{-1}(x))} = \frac{1}{x}$$

* Find the Range of $f(x) = 5\sin^2 x + 2$ is:

$$0 \leq \sin^2 x \leq 1$$
$$0 \leq 5\sin^2 x \leq 5$$
$$2 \leq 2 + 5\sin^2 x \leq 7$$

$R_f = [2, 7]$

* The inverse function of $f(x) = \frac{1}{3 + 3\tan^{-1} x}$

sol:-

$$y = \frac{1}{3 + 3\tan^{-1}(x)} \rightarrow x = \frac{1}{3 + 3\tan^{-1}(y)}$$

$$1 = 3x + 3x\tan^{-1}(y)$$

$$3x\tan^{-1}(y) = 1 - 3x$$

$$\tan^{-1} y = \frac{1 - 3x}{3x} \qquad y = \tan\left(\frac{1-3x}{3x}\right)$$

$\tan x$ $\tan$

الممسوحة ضوئيا بـ CamScanner

* Find Range $f(x) = \frac{6x-2}{3x+1}$

sol:

$$y = \frac{6x-2}{3x+1} \Rightarrow x = \frac{6y-2}{3y+1}$$

$$6y - 2 = 3xy + x$$

$$y(6-3x) = x+2 \rightarrow y = \frac{x+2}{6-3x}$$

$$f^{-1}(x) = \frac{x+2}{6-3x} \qquad D_{f^{-1}(x)} = R_{f(x)}$$

$$D_{f^{-1}} = R - \{2\}$$

---

* Find Domain, $f(x) = 2\sin^{-1}(1-2x)$ is

sol:

$$-1 \geq 1 - 2x \geq +1$$

$$-2 \geq -2x > 0$$

$$1 \leq x \leq 0 \qquad D = [0,1]$$

---

* Find Domain $f(x) = \frac{1}{\frac{2}{x-1} - 1}$

sol:

$$f(x) = \frac{1}{\frac{2}{x-1} - \frac{x-1}{x-1}} \Rightarrow \frac{x-1}{2-x+1}$$

$$f(x) = \frac{x-1}{-x+3} \quad \therefore D = R - \{1, 3\}$$

الممسوحة ضوئيا بـ CamScanner

بسم الله الرحمن الرحيم :

"اللهم اجعل دراستي سهلة وبارك لي في وقتي وجهدي وانفتح لي نجباً يارب العالمين ، اللهم إني استودعك مستقبلي فاسهله خيراً بقدرتك إنك على كل شيء قدير"

* Find Range of $g(x) = -x^2 + 2x - 5$ is

sol :

$$\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right) \Rightarrow -\frac{2}{2 \cdot -1} \rightarrow 1$$

$$f(1) = -1 + 2 - 5 \Rightarrow f(1) = -4$$

Range : $(-\infty, -4]$

---

* $f(x) = 3x - 5$; $(f \circ g)(x) = 12x^2 - 6x + 13$, find $g(c)$??

$\rightarrow f(g(x)) = (f \circ g)(x)$

$\rightarrow 3g(x) - 5 = 12x^2 - 6x + 13$

$\rightarrow 3(g(c)) - 5 = 13$

$\rightarrow 3(g(c)) = 18 \Rightarrow \boxed{g(c) = 6}$ ✱

---

* $f(x) = \frac{5}{x-3}$, $g(x) = x^2 - 1$ Find Domain $(f \circ g)(x)$

$(f \circ g)(x) = D_g \cap D_{f \circ g}$ $f(x^2-1) \rightarrow \frac{5}{x^2-4}$

$= \mathbb{R} \cap \mathbb{R} - \{\pm 2\}$

$= \mathbb{R} - \{\pm 2\}$

---

* $\sin(\tan^{-1}(2))$

sol :-

$= 0.894$

الممسوحة ضوئيا بـ CamScanner

بسم الله الرحمن الرحيم :

اللهم إني أسألك أن تجعلني من دراستي وأن تسهل علي هذا الرئيس

وأن تمنحني قوة الحفظ وتجعلها دليلا على ليلة شاقا ،لا صحبا